



SCENARIO OVERVIEW

Industry	Financial Services
Use Case	Knowledge Management
Cloud Provider	azure
Deployment Model	Cloud-managed

OVERALL ARCHITECTURE SCORE

83/100



Recommended a azure-primary AI Factory for Knowledge Management with a \$500k-\$1M budget. The cloud_managed deployment pattern is optimal for your

INVESTMENT SUMMARY

One-time Setup

\$218K

Monthly Run Rate

\$191K

Annual Run Rate

\$2.5M

3-Year TCO

\$7.1M

BUSINESS CASE

Payback Period	1 months
3-Year Net Value	\$331.8M
NPV (10% discount rate)	\$282.7M
IRR	47.3%
Annual Benefit (total)	\$121.0M

TOP BENEFITS

Employee productivity gains	\$52.3M/yr	HIGH
Process automation savings	\$25.6M/yr	HIGH
Revenue uplift from AI capabilities	\$23.8M/yr	MEDIUM
Infrastructure & ops cost reduction	\$343K/yr	MEDIUM

ARCHITECTURE LAYERS

Cloud Foundation	azure Landing Zone with Enterprise-scale architecture		84
Compute / GPU	NVIDIA DGX H100 SuperPOD (on-prem) + Azure NC-series VMs (A100 / H100) with AKS for burst		83
Storage	Azure Blob Storage + Azure NetApp Files		83
Data Platform	Azure Databricks + Azure Data Factory		85
Model Layer	Azure OpenAI Service (GPT-4o, o1) + Azure ML		82
RAG / Vector Layer	Azure AI Search (integrated vectorization) + Cosmos DB		85

KEY RISKS

- Vendor dependency: azure service changes or pricing revisions may require architectural adjustments — mitigate with open standards and multi-layer abstraction



- Budget variance: GPU compute costs are highly variable at scale — establish reserved instance baselines within 90 days of production launch
- Skills availability: Knowledge Management workloads require specialized ML engineering and MLOps talent — plan hiring or partner engagement in parallel with design
- Data readiness: Data quality, governance, and labelling are the most common blockers to AI factory delivery — begin data assessment in Phase 1
- Security posture: High-sensitivity data classification requires accelerated compliance certification — engage external auditors early

IMPLEMENTATION PHASES

Phase 1: Foundation

4–6 weeks

azure landing zone, enterprise identity (SSO/MFA), hub-and-spoke networking, and security baseline controls

Phase 2: Data & Model Layer

6–10 weeks

Data lakehouse pipeline, vector store population, model API integration, and RAG prototype validation

Phase 3: Orchestration & Applications

6–8 weeks

Agent workflow development, application integration, and human-in-the-loop checkpoint implementation

Phase 4: Observability & Hardening

3–4 weeks

LLM metrics dashboards, security hardening, DR testing, and compliance evidence collection

Phase 5: Scale & Optimize

Ongoing

Reserved instance cost optimization, model fine-tuning, user expansion, and continuous improvement